CHIRAG SHETTY

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PROFESSIONAL SUMMARY

Software Development Engineer with 5+ years of experience building and shipping production generative AI systems at consumer scale – diffusion pipelines, LLM serving, and agentic workflows. Designed and deployed GenAI infrastructure powering India's largest ecommerce platforms (Ajio, JioMart, Tira), integrating Stable Diffusion, Flux and vLLM behind Python (FastAPI) services on GCP with Docker/Kubernetes. Combines generative modelling with deep inference-optimization experience, making large models fast and cheap enough to run in production.

SKILLS

Generative AI	Stable Diffusion, Flux, vLLM, Multimodal LLMs, Super Resolution GAN
LLM Frameworks	LangChain, LangGraph, LlamaIndex, HuggingFace, Agentic Systems, RAG
Languages	Python, C++
Backend	FastAPI, Flask, REST APIs, Authentication & Authorization (JWT/OAuth2)
Model Optimization	TensorRT, ONNX / ONNX Runtime, Post-Training Quantization (PTQ), QAT
Cloud & DevOps	GCP, Docker, Kubernetes, CI/CD Pipelines
Databases	PostgreSQL, SQL
Observability	Logging, Monitoring, Debugging (Production Systems)
Tools	Git, Linux, Cursor, ChatGPT, Claude Code

EXPERIENCE

- **SDE - 2 (ML Engineer) @ Jio Fynd** September 2023 - Present
  Fashion Design Platform (Impetus)

· Architecting a next-generation AI-powered fashion design platform, defining system architecture for automated design generation, variation, and trend exploration.

· Designed agentic workflows using LangChain, LangGraph, and multimodal LLMs to orchestrate complex design workflows; built RAG patterns for grounded design reasoning with curated context, retrieval, context assembly, source-aware prompting, and response evaluation.

· Applied LLM fine-tuning and parameter-efficient adaptation (LoRA/PEFT), with dataset curation and evaluation, balancing model quality, latency, GPU memory, and serving cost.

· Led the integration of multiple generative models (Flux, Stable Diffusion, vLLMs) into a scalable, modular backend architecture.

· Defined design principles focusing on extensibility, performance, and seamless integration with downstream services and product workflows.

GenAI Powered Content Delivery Network (CDN)

· Architected and led the design of a scalable AI-powered CDN serving ecommerce platforms (Ajio, JioMart, Tira), delivering real-time generative image transformations and driving a 5% increase in product engagement.

· Defined and implemented end-to-end generative pipelines for automated product tagging, background generation (Stable Diffusion), foreground extraction (U2Net), and image upscaling (Super Resolution GAN).

· Designed backend architecture and built RESTful APIs using Python (FastAPI) to orchestrate generative workflows, request routing, and integration with platform services.

· Applied Post-Training Quantization (PTQ) to generative and vision models (Stable Diffusion, U2Net, SRGAN), cutting inference latency and GPU memory footprint and producing significant operational cost savings.

· Established system-level optimizations including token usage tracking and pipeline efficiency improvements, significantly reducing operational costs.

- **Machine Learning Engineer @ Adagrad AI** September 2021 - August 2023
  Retail Analytics Platform

· Designed and developed an end-to-end real-time analytics system for retail environments, processing high-volume video streams for footfall and behavior analysis.

· Built high-performance inference pipelines in C++ integrated with deep learning models (YOLO, PyTorch), optimized using NVIDIA TensorRT to achieve 225 FPS on T4 GPUs.

· Applied Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) to compress detection models to INT8 precision, cutting model size and latency while preserving accuracy across distributed deployments.

· Partnered with Fortune 500 clients to deploy solutions globally, adapting systems for diverse hardware and operational constraints.

Automated License Plate Recognition (ALPR)

· Developed a real-time ALPR system deployed on edge devices (NVIDIA Jetson TX2 NX), achieving 50 FPS at 90–95% accuracy across varied vehicle types and real-world conditions.

· Integrated computer vision pipelines with backend services to support enforcement workflows such as restricted lane access (BRT systems).

- **Machine Learning Engineer @ Optisol Business Solutions** November 2020 - September 2021

· Built and deployed ML-driven systems for automating power pole inspections using TensorFlow and OpenCV, enabling scalable defect detection.

· Developed a reinforcement learning (Q-Learning) based system for automating website testing, eliminating manual test-case creation.

· Built backend-integrated workflows using FastAPI/Flask to orchestrate automated pipelines and improve deployment efficiency.

PROJECTS

· **Open Source Contribution - BYTETrack: Multi-Object Tracking**

· BYTETrack is a widely-used Multi-Object Tracking (MOT) algorithm used in conjunction with object detection. Implemented and contributed the C++ port of the tracking algorithm and integrated it with the NVIDIA DeepStream SDK, enabling real-time, GPU-accelerated multi-object tracking for production video analytics pipelines.

· **Shop the Look - Fashion Recommendation Engine**

· A fashion recommendation engine that utilizes image similarity, ranking and search to provide personalized, content-based suggestions.

· **Mercedes Benz Greener Manufacturing - Kaggle**

· Kaggle problem for estimating the bench time of automobiles. Achieved KPI score that was ranked within the top 100 submissions on Kaggle leader-board.

· **Clustering Locations - Unsupervised Machine Learning**

· Unsupervised locality clustering for optimising business decisions. Used data mining and web scraping for creating relevant features to solve the problem.

EDUCATION

Vellore Institute of Technology, Vellore 2016 - 2020
Bachelor of Technology, Mechanical Engineering CGPA - 8.33/10